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SUBSTITUTED EPOXIDE MODIFIED SORBENTS, SYSTEMS INCLUDING SORBENTS, AND METHODS USING THE SORBENTS

Abstract

The present disclosure provides for substituted epoxide modified sorbents and contactors, methods of using sorbents and contactors to capture CO_2, structures including the sorbent, and systems and devices using sorbents and contactors to capture CO_2. The present disclosure provides for sorbents and contactors that can include a CO_2-philic phase and a support. The CO_2-philic phase can include a modified amine polymer that is the reaction product of an amine and a substituted epoxide.

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Background/Summary

CLAIM OF PRIORITY TO RELATED APPLICATION [0001] This application claims priority to co-pending U.S. provisional application entitled "SORBENTS, SYSTEMS INCLUDING SORBENTS, AND METHODS USING THE SORBENTS" having Ser. No. 63/364,308 filed on May 6, 2022, which is entirely incorporated herein by reference. In addition, this application claims priority to co-pending U.S. provisional application entitled "SORBENTS, SYSTEMS INCLUDING SORBENTS, AND METHODS USING THE SORBENTS" having Ser. No. 63/431,512 filed on Dec. 9, 2022, which is entirely incorporated herein by reference.

BACKGROUND
[0002] Greenhouse gases trap heat in the atmosphere and carbon dioxide (CO.sub.2) is one of the main greenhouse gases. CO.sub.2 is emitted through human related activities such as transportation, electric power, industry and agriculture. In particular, CO.sub.2 emissions are caused by burning fossil fuels, solid waste, and trees as well as through the manufacture of cement and other materials. One way to decrease the amount of CO.sub.2 in the atmosphere is to capture CO.sub.2 using materials having an affinity for CO.sub.2. There is a need for materials that can effectively capture CO.sub.2.
SUMMARY
[0003] The present disclosure provides for substituted epoxide modified sorbents and contactors, methods of using sorbents and contactors to capture CO.sub.2, structures including the sorbent, and systems and devices using sorbents and contactors to capture CO.sub.2.
[0004] An aspect of the present disclosure provides for a sorbent comprising: a CO.sub.2-philic phase and a support, wherein the CO.sub.2-philic phase includes the reaction product of a substituted epoxide and an amine.
[0005] An aspect of the present disclosure provides for a contactor, comprising a structure and the sorbent as described above and herein.
[0006] An aspect of the present disclosure provides for a system for capturing CO.sub.2 from a gas, optionally the gas is ambient air, comprising: a first device configured to introduce the gas to the sorbent or contactor as described above and herein to bind CO.sub.2 to the sorbent; a second device configured to heat the sorbent containing bound CO.sub.2 to at least a first temperature to release the CO.sub.2; and a third device configured to collect the released CO.sub.2.
[0007] An aspect of the present disclosure provides for a method of capturing CO.sub.2 from gas optionally the gas is ambient air comprising: introducing the ambient air to the sorbent as described above and herein to bind CO.sub.2 to the sorbent; heating the sorbent to at least a first temperature to controllably release the CO.sub.2; and collecting the CO.sub.2 in a CO.sub.2 collection device.
[0008] An aspect of the present disclosure provides for a system for implement the method as described above and herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Further aspects of the present disclosure will be more readily appreciated upon review of the detailed description of its various embodiments, described below, when taken in conjunction with the accompanying drawings.

[0010] FIG. **1**A shows a schematic of how amine moieties bind CO_2 into the form of a carbamate. FIG. **1**B is a schematic of a sorbent system comprised of support and CO_2-philic phase. Together, the support and CO_2-philic phase comprise a sorbent.

[0011] FIG. **2** is a schematic of honeycomb monolith contactor comprised of a substrate and a sorbent washcoat.

[0012] FIG. **3** illustrates substituted epoxides evaluated as modifiers to polyethylenimine (PEI).

[0013] FIG. **4** illustrates mass loss curves during exposure of sorbents created with improved CO_2-philic phases containing PEI reacted with substituted epoxides and unmodified PEI to diluted air while ramping the temperature from room temperature to 900° C.

[0014] FIG. **5** illustrates heat flow (DSC) curves during exposure of sorbents created with improved CO_2-philic phases containing PEI reacted with substituted epoxides and unmodified PEI to diluted air while ramping the temperature from room temperature to 900° C.

[0015] FIG. **6** illustrates transient mass change profiles from TGA CO_2 uptake experiments at 400 ppm CO_2 (DAC conditions) utilizing improved sorbents containing PEI reacted with substituted epoxides; each at 0.5 mol substituted epoxide/mol primary N in PEI, and all supported in a mesoporous alumina. Data for a sorbent utilizing PEI is shown for reference as well. Data are reported as amine efficiency (mol CO_2/mol N in PEI present in sample).

[0016] FIGS. **7**A-**7**D illustrate extent of oxidation over time of PEI, determined via the differential scanning calorimetry method described herein and discussed in the referenced publication (solid lines, DSC), and via the loss of amine efficiency (datapoints, AE) at (FIG. **7**A) 5%, (FIG. **7**B) 17%, and (FIG. **7**C) 30% O_2 concentration; (FIG. **7**D) extent of oxidation with different PEI pore fillings. Figure taken from Nezam et al, ACS Sustainable Chem. Eng., 2021, 9, 8477-8486.

[0017] FIG. **8**: Transient oxidation curves for PEI and PEI reacted with substituted epoxides tested under 17% O_2, balance N_2 at 137.5° C., all sorbents are in a mesoporous alumina, substituted epoxide ratios are 0.5 mol substituted epoxide/mol primary N in PEI.

[0018] FIG. **9** illustrates illustrative example structures of modified amines on modified amine polymers.

[0019] FIG. **10** illustrates example chemical structures of 1-amino-2-alcohol groups reacted with a primary amine to form a secondary amine.

DETAILED DISCLOSURE

[0020] Embodiments of the present disclosure provide for substituted epoxide modified sorbents (also referred to herein as “sorbent” or “sorbents”) and contactors, methods of using sorbents and contactors to capture CO_2, structures including the sorbent, and systems and devices using the sorbents and contactors to capture CO_2. The methods, systems, and sorbents of the present disclosure can be advantageous over current technologies since they are relatively more robust and reduce the cost of capturing CO_2, in particular from ambient air. In an aspect, the present disclosure provides for sorbents and contactors having an improved CO_2-philic phase that has a greater resistance to oxidation. In an aspect, the CO_2-philic phase is created by the modification of an amine with a reactive molecule such as to form a modified amine.

[0021] Before the present disclosure is described in greater detail, it is to be understood that this disclosure is not limited to particular embodiments described, and as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the present disclosure will

be limited only by the appended claims.
[0022] Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the disclosure. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges and are also encompassed within the disclosure, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the disclosure.
[0023] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. Although any methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the present disclosure, the preferred methods and materials are now described.
[0024] As will be apparent to those of skill in the art upon reading this disclosure, each of the individual embodiments described and illustrated herein has discrete components and features which may be readily separated from or combined with the features of any of the other several embodiments without departing from the scope or spirit of the present disclosure. Any recited method can be carried out in the order of events recited or in any other order that is logically possible.
[0025] Embodiments of the present disclosure will employ, unless otherwise indicated, techniques of chemistry, materials science, mechanical engineering, and the like, which are within the skill of the art.
[0026] The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how to perform the methods and use the probes disclosed and claimed herein. Efforts have been made to ensure accuracy with respect to numbers (e.g., amounts, temperature, etc.), but some errors and deviations should be accounted for. Unless indicated otherwise, parts are parts by volume, temperature is in ° C., and pressure is at or near atmospheric. Standard temperature and pressure are defined as 20° C. and 1 atmosphere.
[0027] Before the embodiments of the present disclosure are described in detail, it is to be understood that, unless otherwise indicated, the present disclosure is not limited to particular materials, reagents, reaction materials, manufacturing processes, or the like, as such can vary. It is also to be understood that the terminology used herein is for purposes of describing particular embodiments only, and is not intended to be limiting. It is also possible in the present disclosure that steps can be executed in different sequences where this is logically possible.
[0028] It must be noted that, as used in the specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a compound” includes a plurality of compounds. In this specification and in the claims that follow, reference will be made to a number of terms that shall be defined to have the following meanings unless a contrary intention is apparent.
Definitions
[0029] By “chemically feasible” is meant a bonding arrangement or a compound where the generally understood rules of organic structure are not violated. The structures disclosed herein, in all of their embodiments are intended to include only “chemically feasible” structures, and any recited structures that are not chemically feasible, for example in a structure shown with variable atoms or groups, are not intended to be disclosed or claimed herein. However, if a bond appears to be intended and needs the removal of a group such as a hydrogen from a carbon, the one of skill would understand that a hydrogen could be removed to form the desired bond.
[0030] The term “substituted” refers to any one or more hydrogen atoms on the designated atom (e.g., a carbon atom) that can be replaced with a selection from the indicated group (e.g., halide,

hydroxyl, alkyl, and the like), provided that the designated atom's normal valence is not exceeded. As used herein, the term “optionally substituted” typically refers to from zero to four substituents, wherein the substituents are each independently selected. Each of the independently selected substituents may be the same or different than other substituents. For example, the substituents (e.g., an R type group) of a formula may be optionally substituted (e.g., from 1 to 4 times) with independently selected H, halogen, hydroxy, acyl, alkyl, alkenyl, alkynyl, cycloalkyl, heterocyclo, aryl, heteroaryl, alkoxy, amino, amide, thiol, sulfone, sulfoxide, oxo, oxy, nitro, carbonyl, carboxy, amino acid sidechain, amino acid, etc. In an embodiment, substituted includes substitution with a halogen. Each group (e.g., alkyl, allyl, phenyl, phenol, benzyl, nitroimidazole, aryl, heteroaryl, alkoxy, alkyl amino,) can be substituted or unsubstituted.

[0031] As used herein, a “derivative” of a compound refers to a chemical compound that may be produced from another compound of similar structure in one or more steps, as in replacement of H by an alkyl, acyl, amino group, etc.

[0032] As used herein, “alkyl” or “alkyl group” refers to a saturated aliphatic hydrocarbon, which can be straight or branched, having 1 to 40, 1 to 20, 1 to 10, or 1 to 5 carbon atoms, where the stated range of carbon atoms includes each intervening integer individually, as well as sub-ranges. Examples of alkyl groups include, but are not limited to methyl, ethyl, n-propyl, i-propyl, n-butyl, s-butyl, t-butyl, n-pentyl, and s-pentyl. Reference to “alkyl” or “alkyl group” includes unsubstituted and substituted forms of the hydrocarbon moiety.

[0033] As used herein, “halo”, “halogen”, “halide”, or “halogen radical” refers to a fluorine, chlorine, bromine, iodine, and astatine, and radicals thereof. Further, when used in compound words, such as “haloalkyl” refers to an alkyl or alkenyl radical in which one or more hydrogens are substituted by halogen radicals.

[0034] As used herein an “allyl” group refers to —CH.sub.2—HC═CH.sub.2.

[0035] As used herein a “phenyl” group refers to a 6 carbon aromatic cycle (C.sub.6H.sub.5—) that can be bonded through the carbon without the hydrogen as a functional group in a compound.

[0036] As used herein a “phenol” group refers to a 6 carbon aromatic cycle (C.sub.6H.sub.5OH—) that can be bonded through the carbon without the hydrogen as a functional group in a compound.

[0037] As used herein a “benzyl” group refers to a 6 carbon aromatic cycle where one of the hydrogens on the ring is replaced with —C(H.sub.2)— and —C(H.sub.2)— can be bonded through the carbon as a functional group in a compound (ring —C(H.sub.2)—X).

[0038] As used herein a “nitroimidazole” group refers to (O.sub.2NC.sub.3H.sub.2N.sub.2H) or
##STR00001##
that can be bonded through the nitrogen to replace the hydrogen in the heterocyclic ring.

[0039] As used herein an “alkyl amino” group refers to (R.sub.1—NH.sub.2 or R.sub.1—NH—R.sub.2). Optionally each R can be an —(CH.sub.x).sub.y or H), that can be bonded through the carbon to replace a hydrogen, where x can be 2 or 3 and y can be 1 to 20.

[0040] “Aryl”, as used herein, refers to C.sub.5-C.sub.10-membered aromatic, heterocyclic, fused aromatic, fused heterocyclic, biaromatic, or biheterocyclic ring systems. Broadly defined, “aryl”, as used herein, includes 5-, 6-, 7-, 8-, 9-, and 10-membered single-ring aromatic groups that may include from zero to four heteroatoms, for example, benzene, pyrrole, furan, thiophene, imidazole, oxazole, thiazole, triazole, pyrazole, pyridine, pyrazine, pyridazine and pyrimidine, and the like.

[0041] Those aryl groups having heteroatoms in the ring structure may also be referred to as “heteroaryl” or “heteroaromatics”.

[0042] “Heteroaryl” refers to any stable 5, 6, 7, 8, 9, 10, 11, or 12 membered, (unless the number of members is otherwise recited), monocyclic, bicyclic, or tricyclic heterocyclic ring that is aromatic, and which consists of carbon atoms and 1, 2, 3, or 4 heteroatoms independently selected from the group consisting of N, O, and S. If the heteroaryl is defined by the number of carbons atoms, then 1, 2, 3, or 4 of the listed carbon atoms are replaced by a heteroatom. If the heteroaryl group is bicyclic or tricyclic, then at least one of the two or three rings must contain a heteroatom, though both or all three may each contain one or more heteroatoms. If the heteroaryl group is bicyclic or tricyclic, then only one of the rings must be aromatic. The N group may be N, NH, or N-substituent, depending on the chosen ring and if substituents are recited. The nitrogen and sulfur heteroatoms may optionally be oxidized (e.g., S, S(O), $S(O)_2$, and N—O). The heteroaryl ring may be attached to its pendant group at any heteroatom or carbon atom that results in a stable structure. The heteroaryl rings described herein may be substituted on carbon or on a nitrogen atom if the resulting compound is stable.

[0043] The term “heteroatom” means for example oxygen, sulfur, nitrogen, phosphorus, or silicon (including, any oxidized form of nitrogen, sulfur, phosphorus, or silicon; the quaternized form of any basic nitrogen or; a substitutable nitrogen of a heterocyclic ring).

[0044] The term “alkoxy” represents an alkyl group as defined above with the indicated number of carbon atoms attached through an oxygen bridge. Examples of alkoxy include, but are not limited to, methoxy, ethoxy, n-propoxy, i-propoxy, n-butoxy, s-butoxy, t-butoxy, n-pentoxy, and s-pentoxy. The term “lower alkoxy” means an alkoxy group having less than 10 carbon atoms.

DISCUSSION

[0045] The present disclosure provides for substituted epoxide modified sorbents (also referred to herein as “sorbent” or “sorbents”) and contactors, methods of using sorbents and contactors to capture CO_2, structures including the sorbent, and systems and devices using sorbents and contactors to capture CO_2. The present disclosure is directed to multiple types of sorbents and structures that will be described below and herein.

[0046] In an aspect, the present disclosure provides for sorbents and contactors that include a CO_2-philic phase and a support, where the CO_2-philic phase includes a modified amine polymer that is the reaction product of an amine and a substituted epoxide. The reaction product is also described herein. The CO_2-philic phase includes CO_2 binding molecules The CO_2 binding molecules contain CO_2 binding moieties. In an aspect, the modified amine polymer maintains high CO_2 capacities relative to the unmodified amine polymer. In an aspect, the sorbents with the modified amine polymer lose less of the capacity to capture CO_2 following oxidative exposures compared to sorbents created with the unmodified amine polymer, thereby creating an improved CO_2-philic phase with a longer commercial lifetime relative to CO_2-philic phase without epoxide.

[0047] A structured support, also referred to as a formed support or as a structure, refers to a support that has been formed into a structure where the structure is, at standard conditions, a solid body. Supports can also be unstructured, at standard conditions having a powdery consistency. When a support is referred to without mention to a structure, forming, or being formed or structured, it can refer to supports that are either structured or unstructured.

[0048] Structured supports can take the form of a homogeneous solid body (i.e., comprised predominantly of support but also containing components that allow it to remain a stable body at standard conditions) or as a coating on a substrate, whereby the substrate has a different composition than the coating and provides the mechanical stability to the coating.

[0049] It can be useful to utilize a structured support with a CO_2-philic phase as a contactor in a process for removing CO_2 from a gas stream such as ambient air. Contactors provide a geometry to a CO_2-philic phase such that considerations such as pressure drop, throughput, and/or mass transfer rates can be optimized.

[0050] The CO_2 binding molecules can be an amine or an amine polymer. The amine or amine polymer can contain primary amines, secondary amines, tertiary amines, or a mixture of any combination of primary, secondary, and tertiary amines. The amine polymer can be branched, hyperbranched, dendritic, or linear. The CO_2 binding moieties are the amine moieties on the amine molecule or polymer. The amine moieties can interact with CO_2 to form carbamate, carbonate, or bicarbonate species. FIG. **1**A shows a schematic of how amine moieties bind CO_2 into the form of a carbamate.

[0051] Primary amines are defined as having the chemical structure —NH.sub.2R.sup.1, where R.sup.1 is an alkyl group such as CH.sub.2 or CH.sub.3. Secondary amines are defined as having the chemical structure —NHR.sup.1R.sup.2, where R.sup.1 and R.sup.2 are independently selected from an alkyl group such as CH.sub.2 or CH.sub.3. Tertiary amines are defined as having the chemical structure —NR.sup.1R.sup.2R.sup.3, where R.sup.1, R.sup.2, and R.sup.3 are independently selected from an alkyl group such as CH.sub.2 or CH.sub.3.
[0052] Linear amine polymers can be defined as containing only primary amines, secondary amines, or both primary and secondary amines. The ratio of secondary to primary amines can be about 0.5 to 10,000. In an aspect, the linear amine polymer can have a molecular weight of about 100 to 100,000 g/mol, about 200 to 30,000 g/mol or about 600 to 5,000 g/mol.
[0053] Branched amine polymers can be defined as containing any number of primary, secondary, and tertiary amines, which does not overlap linear amine polymers or dendritic amine polymers. The ratio of primary, secondary, and tertiary can be about 10:80:10 to 60:10:30, about 60:30:10 to 30:50:20, or about 45:45:10 to 35:45:20. As one of skill would understand, the chemical structures of branched amine polymer can vary greatly and can be very complex. In an aspect, the branched amine polymer can have a molecular weight of about 100 to 100,000 g/mol, about 200 to 30,000 g/mol or about 600 to 5,000 g/mol.
[0054] Dendritic amine polymers can be defined as containing only primary and tertiary amines, where groups of repeat units are arranged in a manner that is necessarily symmetric in at least one plane through the center (core) of the molecule, where each polymer branch is terminated by a primary amine, and where each branching point is a tertiary amine. The core or central linkage is the same as the branching amines (e.g., ethylenimine core and ethylenimine branches, propylenimine core and propylenimine branches). The ratio of primary to tertiary can be about 1 to 3. In an aspect, the dendritic amine polymer can have a molecular weight of about 100 to 100,000 g/mol, about 200 to 30,000 g/mol or about 280 to 3,000.
[0055] Hyperbranched amine polymers can be defined as having chemical structure resembling dendritic amine polymer, but containing defects in the form of secondary amines (e.g., linear subsections as would exist in a branched polymer), in such a way that provides a random chemical structure instead of a symmetric chemical structure. The hyperbranched amine polymers do not overlap branch amine polymers or dendritic polymers. In a hyperbranched chemical structure, the ratio of primary to secondary to tertiary can be about 65:5:30 to 30:10:60. In an aspect, the hyperbranched amine polymer can have a molecular weight of about 100 to 100,000 g/mol, about 200 to 30,000 g/mol or about 600 to 10,000 g/mol.
[0056] In an aspect, linear, hyperbranched and branched amine polymers have secondary amines and dendritic amines do not, which may be advantageous since secondary amines bond strongly to CO.sub.2.
[0057] In an aspect, the amine polymer can be polyethylenimine, polypropylenimine, polyallylamine, polyvinylamine, polyglycidylamine, polystyrene-divinylbenzene polymer functionalized with amine such as alkylbenzylamine moieties, or other amine polymer, where each can be branched, hyperbranched, dendritic, or linear.
[0058] In an embodiment, the size (e.g., length, molecular weight), amount (e.g., number of distinct amine polymers), and/or type of amine polymer, can be selected based on the desired characteristics of the porous structure (e.g., CO.sub.2 absorption, regenerative properties, oxidative stability, loading, and the like).
[0059] In an aspect, the modified amine polymer can contain primary amines, secondary amines, tertiary amines or a mixture of any combination of primary, secondary, and tertiary amines, each of which is defined above. In an aspect, the amine can be a primary amine and the substituted epoxide can be an alkyl (e.g., linear or branched C1-C20 or linear or branched C1-C10) or cyclic epoxide. In another aspect, the amine can be a secondary amine and the substituted epoxide can be an alkyl (e.g., linear or branched C1-C20 or linear or branched C1-C10) or cyclic epoxide. The modified

amine polymer can be branched, hyperbranched, dendritic, or linear, each of which is defined above.

[0060] In an aspect, the modified amine polymer can be primary or secondary before the reaction with the substituted epoxide, such as to form a secondary or tertiary amine. Illustrative structures are shown below, where R.sub.x is the substituted group:

##STR00002##

In an aspect, R.sub.1, R.sub.2, R.sub.3, and R.sub.4 can be a hydrogen atom, an alkyl group, an allyl group, an alkoxy group, an alkylhalide group, an aryl group, a benzyl group, a phenyl group, a phenol group, a heteroaryl group, a nitroimidazole group and the like, and combinations thereof (each group can independently be substituted or unsubstituted). In another aspect, R.sub.1, R.sub.2, R.sub.3, and R.sub.4 can be a hydrogen atom, an alkyl group, an allyl group, an alkylhalide group, an aryl group, a benzyl group, a phenyl group, a phenol group, a heteroaryl group, a nitroimidazole group and the like, and combinations thereof (each group can independently be substituted or unsubstituted). In an aspect, each R′ can be independently selected from a hydrogen atom, an alkyl group, an alkoxy group, an alkylhalide group, an aryl group, a benzyl group, a phenyl group, a phenol group, an amine group (e.g., alkylenimine (C2 to C8) such as ethylenimine and propylenimine), a heteroaryl group, a nitroimidazole group, an alkyl amino group, and the like, and combinations thereof (each group can independently be substituted or unsubstituted). In another aspect, each R′ can be independently selected from a hydrogen atom, an alkyl group, an alkylhalide group, an aryl group, a benzyl group, a phenyl group, a phenol group, an amine group (e.g., alkylenimine (C2 to C8) such as ethylenimine and propylenimine), a heteroaryl group, a nitroimidazole group, an alkyl amino group, and the like, and combinations thereof (each group can independently be substituted or unsubstituted).

[0061] In an aspect, the modified amine polymer can be a modified polyethylenimine, a modified polypropylenimine, a modified polyallylamine, a modified polyvinylamine, a modified polyglycidylamine, a modified polystyrene-divinylbenzene polymer functionalized with amine such as alkylbenzylamine moieties, or other modified amine polymers, where in each the modified amine polymer can be branched, hyperbranched, dendritic, or linear, each of which is defined above and herein.

[0062] In an aspect, the fraction of amines modified according to the description herein can be 0.001 to 1 or 0.01 to 1 or 0.1 to 1 or 0.5 to 1 of total primary and secondary amines in the amine polymer, where a fraction of 1 means all of the primary and secondary amines, or a fraction of about 0.01 to 0.5 amines in the amine polymer.

[0063] In an aspect, the CO.sub.2-philic phase includes a modified amine polymer that is the reaction product of an amine and a substituted epoxide. The substituted epoxide refers to an epoxide having one or more substituent groups. Also, it is not an epoxide with a single linear aliphatic group substituent. For example, the substituent groups can include an alkyl group, an allyl group, an alkylhalide group, an aryl group, a benzyl group, a phenyl group, a phenol group, a heteroaryl group, a nitroimidazole group and the like (each group can independently be substituted or unsubstituted). For example, the substituted epoxide that reacts with an amine can include one or more of the following: 1,2-epoxy-2-methylpropane, 2,3-epoxybutane, 3,4-epoxy-1-butene, 3,3-dimethyl-1,2-epoxybutane, 2-methyl-2-vinyloxirane, 1,2-epoxy-3,3,3-trifluoropropane, 2-phenylpropylene oxide, 1-(2,3-epoxypropyl)-2-nitroimidazole, stilbene oxide (e.g., trans-stilbene oxide, cis-stilbene oxide, or a mixture of trans-stilbene oxide, cis-stilbene oxide) 1,2-epoxy-5-hexene, epoxystyrene, (2,3-epoxypropyl)benzene, 2-(4-chlorophenyl) oxirane, 2-(4-bromophenyl) oxirane, 2-(4-fluorophenyl) oxirane, cyclohexene oxide, 4-vinyl-1-cyclohexene-1,2-epoxide, and 1,3-butadiene diepoxide.

[0064] In an aspect, the modified amine polymer can have a 1-amino-2-alcohol structure in addition to having the substituted group. While a particular embodiment of the modified amine polymer can be referred to a 1-amino-2-alkylalcohol, an alternative nomenclature can be 1,2-

alkanolamine, regardless the structure is illustrated herein. The modified amine polymer can have a 1-amino alkyl structure in addition to the substituted group, where the alkyl structure can be linear and have between 1 to 5 carbons in the carbon chain. The amine can be primary or secondary before the reaction, such as to form a secondary or tertiary amine. Illustrative structures are shown below, where R is the substituted group:
##STR00003##
In an aspect, R can be an alkyl chain (e.g., C2 to C12), an aryl group, a benzyl group, a phenyl group, a phenol group, a heteroaryl group and the like, and combinations thereof (each group can independently be substituted or unsubstituted).

[0065] Although not intending to be bound by theory, the CO.sub.2-philic phase modified by the reaction with the substituted epoxide allows for the ability to tune the properties of the CO.sub.2-philic phase. For example, bulky substituent groups such as phenyl or substituted phenyl can be used to sterically impede the oxygen attack during an oxidation reaction. Additionally, substituted groups can also be used to chemically stabilize the amine polymer from oxidation. Chemical stabilization can be achieved by the introduction of electron donating or withdrawing substituent groups onto the amine polymer. The amount, type, and mixture quantity of substituted epoxide reacted with the CO.sub.2-philic phase can be tuned and changed to achieve the desired properties of the CO.sub.2-philic phase. Tuning the CO.sub.2-philic phase can result in one or a combination of the following that yield an improved CO.sub.2 sorbent: increased lifetime due to a reduction in the rate of oxidative degradation, increase in amine efficiency of the sorbent, increase in the CO.sub.2 swing capacity of the sorbent in an adsorption/desorption process, increase in the equilibrium capacity of the sorbent.

[0066] In an aspect, it can be advantageous to improve the stability of the CO.sub.2-philic phase to process conditions relevant to use of the sorbent in a CO.sub.2 separation process, particularly during sorbent regeneration (process cycles that raise the temperature of the sorbent to remove bound CO.sub.2). It can also be advantageous to improve the stability of the CO.sub.2-philic phase to conditions relevant to storage of the sorbents when they are not being utilized in a process or plant. Sorbents that have a CO.sub.2-philic phase with improved stability with respect to process conditions including sorbent regeneration, storage, or both process conditions including sorbent regeneration and storage are valuable.

[0067] Evaluating the oxidative stability of materials in environments that contain oxygen and CO.sub.2 is useful due to the fact that during regeneration processes, desorbed CO.sub.2 is present at different concentrations in addition to oxygen at elevated temperatures, and can impact the stability of the material. Separately, evaluating the oxidative stability of materials with oxygen only (CO.sub.2-free air) is a useful way to evaluate the shelf life of a material when it is stored at ambient conditions.

[0068] As described above, the CO.sub.2-philic phase (e.g., the CO.sub.2 binding molecules) can be homogeneous or heterogeneous. When the CO.sub.2-philic phase is heterogeneous, the CO.sub.2 binding molecules can be present in a variety of ways. For example, the CO.sub.2 binding molecules can be applied or incorporated to form a layer of the CO.sub.2-philic phase on a support, such as on the surface of pores of the support. In another aspect, independent of or used in combination with other aspects such as those described above, the CO.sub.2 binding molecules can be used to form a part of or all of the support, where CO.sub.2-philic phase functions as described herein. Various combinations are contemplated and are part of the present disclosure. Additional ways in which to apply, use, or incorporate the CO.sub.2-philic phase homogeneously and/or heterogeneously are described herein and below.

[0069] As described herein, the sorbent includes the CO.sub.2-philic phase (e.g., the CO.sub.2 binding molecules) and the support. The support includes a surface (e.g., a surface that can be exposed to a gas including CO.sub.2 during regular use and/or that can interact with the CO.sub.2-philic phase). The surface can be the surface of pores and/or other surfaces that the CO.sub.2-philic

phase contacts or interacts with.
[0070] In an aspect, the CO.sub.2-philic phase (e.g., the CO.sub.2 binding molecules) can be disposed on and/or within a support to form a sorbent. The CO.sub.2-philic phase can be disposed on the surface of the support, and/or within pores of the support, and/or on exterior surface of a support or any combination thereof. In an aspect, the CO.sub.2-philic phase can be a coating on the surface of the porous material, a monolayer on the surface of the porous material, a self-assembled monolayer on the surface of the porous material, a bulk phase within the pores of the porous material, a coating on the exterior surface of the porous material, and the like.
[0071] In an aspect, the support can be made of one or more types of materials such as ceramic, metal, metal oxide, plastic, cellulose, carbon, a zeolite, a metal organic framework (MOF), a porous organic framework (POF), a covenant organic framework (COF), a polymer of intrinsic microporosity (PIM), a polymer, a fibrous cellulose, fiberglass, boron-nitride fiber, and the like. In another aspect, the support can be made of materials that also include the CO.sub.2-philic phase.
[0072] The metal oxide support can be selected from cordierite, alumina (e.g., γ-alumina, θ-alumina, δ-alumina), cordierite-a-alumina, silica, aluminosilicates, zirconia, germania, magnesia, titania, hafnia, silicon nitride, zircon mullite, spodumene, alumina-silica magnesia, zircon silicate, sillimanite, magnesium silicates, zircon, petalite, and combinations thereof. In cases where the oxide contains a formal charge, the charge can be balanced by appropriate counter-ions, such as cations of NR.sub.4, Na, K, Ca, Mg, Li, H, Rb, Sr, Ba, Cs or anions including phosphate, phosphite, sulfate, sulfate, nitrate, nitrite, chloride, bromide and the like. The metal oxide can contain dopants such as zirconium, iron, tin, silicon, titanium, magnesium, and combinations thereof. It is known that metal oxides can contain acid, base, and neutral sites on their surfaces and that dopants can alter the amount and strength of acid and base sites on the surfaces.
[0073] In an aspect, the polymer support can be a polymer and/or copolymer of polyolefin(s), polyester(s), polyurethane(s), polycarbonate(s), polyetheretherketone(s), polyphenylene oxide(s), polyether sulfone(s), melamine(s), polyamide(s), polyvinylbenzene, polystyrene-divinylbenzene, polyurethane, polyacrylates, polystyrenes, polyacrylonitriles, polyimides, poly furfural alcohol, phenol furfuryl alcohol, melamine formaldehydes, resorcinol formaldehydes, cresol formaldehyde, phenol formaldehyde, polyvinyl alcohol dialdehyde, polycyanurates, polyacrylamides, various epoxies, agar, and agarose, or combinations thereof.
[0074] The support can be porous (e.g., macroporous, mesoporous, microporous, or mixtures thereof (e.g., where a macroporous surface can include mesopores, and/or micropores, within one or more of the macropores, where a mesoporous surface can include micropores, and so on)). In an aspect, the porous structure is mesoporous. The pores can extend through the porous structure or porous layer or only extend to a certain depth. The macropores of the porous structure can have pores having a diameter of about 100 nm to 10,000 nm, a length of about 500 nm to 100,000 nm and a volume of 0.2-1 cc/g. The mesopores of the porous structure can have pores having a diameter of about 5 nm to 100 nm, a length of about 10 nm to 10,000 nm, and a volume of 0.1-2 cc/g. The micropores of the porous structure can have pores having a diameter of about 0.5 to 5 nm, a length of about 0.5 nm to 1000 nm and a volume of about 0.1-1 cc/g.
[0075] The support can be porous and have a porosity of at least 90%, at least 85%, at least 80%, at least 75%, at least 70%, at least 65%, at least 60%, or about 60 to 90%. In some embodiments, the support can have a surface area of 1 m.sup.2/g or more, 10 m.sup.2/g or more, 100 m.sup.2/g or more, 150 m.sup.2/g or more, 200 m.sup.2/g or more, or 250 m.sup.2/g or more, 500 m.sup.2/g or more, 1000 m.sup.2/g or more.
[0076] In an embodiment, the CO.sub.2-philic phase (e.g., the CO.sub.2 binding molecules) can be physically impregnated in the internal volume pores of the porous structure and not covalently bonded to the internal surface of the pores of the porous structure, can be grafted (e.g., covalently bonded directly or indirectly) to the internal surface of the pores of the porous structure, or a combination thereof. In an embodiment, the CO.sub.2-philic phase (e.g., the CO.sub.2 binding

molecules) can be covalently bonded (e.g., directly to the surface or via a linker group) to the surface of the material, which may include the internal surface of pores for a porous layer or porous structure. In an aspect, the covalent bonding can be achieved using known techniques in the art for bonding sorbents. In regard to the CO.sub.2-philic phase (e.g., the CO.sub.2 binding molecules) being physically impregnated in the pores of the porous structure and not covalently bonded to the internal surface of the pores of the porous structure, the CO.sub.2-philic phase can be confined within the pores of the support, but not bonded to the surface. In yet another embodiment, the CO.sub.2-philic phase (e.g., the CO.sub.2 binding molecules) is present in a plurality of pores (internal volume) of the porous structure ("porous structure" can include a structure having pores in its surface or a structure having a porous layer or coating on the surface of the structure (where the structure itself may or may not be porous)), where the CO.sub.2-philic phase has a loading of about 10% to 75% by weight of the support. In regard to the loading, the loading is determined by thermogravimetric analysis (TGA).

[0077] In an embodiment, the support can include a surface layer on the surface of the pores of the support that can bond with the CO.sub.2-philic phase. In an aspect, the surface layer can include organically modified moieties (e.g., alkyl groups, amines, thiols, phosphines, and the like) on the surface (e.g., outside and/or inside surfaces of pores) of the material. In an embodiment, the surface layer can include surface alkyl groups, amines, thiols, phosphines, and the like, that the CO.sub.2-philic phase can directly covalently bond and/or indirectly covalently bond (e.g., covalently bond to a linker covalently bonded to the material). In an embodiment, the surface layer can include an organic polymer having one or more of the following groups: alkyl groups, amines, thiols, phosphines, and the like. In another embodiment, the structure can be a carbon support, where the carbon support can include one or more of the following groups: alkyl groups, amines, thiols, phosphines, and the like.

[0078] In an embodiment, the CO.sub.2-philic phase can occupy about 10 to 100% of the mesopore volume of the support, or can occupy about 30 to 90% of the mesopore volume of the support, or can occupy about 40 to 80% of the mesopore volume of the support, or can occupy about 50 to 70% of the mesopore volume of the support.

[0079] The process of making a formed support or structure described above and herein can be used to create any of the structures listed in this paragraph and those in the following paragraphs. The sorbent, comprising the CO.sub.2-philic phase and the support, can be formed into or applied to a structure. In an aspect, the CO.sub.2-philic phase and the support can form 100% of the structure or less than 100% (e.g., each combination of ranges between about 10%, about 20%, about 30%, about 40%, about 50% and about 60%, about 70%, about 80%, about 90%, about 99% such as about 10-99%, about 10 to 80%, about 10 to 50%, about 50 to 99%, about 50 to 90%, about 50 to 80%), where a sufficient amount of sorbent is on the surface of the structure to absorb the desired amount of CO.sub.2. In an aspect, the structure can be a honeycomb, a laminate sheet, a foam, fibers, minimal surface solids, powder trays, pellets, powder, and the like or a combination of two or more of the foregoing.

[0080] In an aspect, the structure can be porous and have a porosity of at least 90%, at least 85%, at least 80%, at least 75%, at least 70%, at least 65%, at least 60%, or about 60 to 90%.

[0081] In an aspect, the porosity of the structure can be comprised of macropores, mesopores, and/or micropores. In an aspect, the CO.sub.2-philic phase is predominantly (e.g., about 40 to 100% or about 50 to 90%, or about 60 to 80%) located within the mesopores of the structure.

[0082] In an aspect, the structure can be comprised entirely of sorbent or can contain another substrate material such as ceramic, metal, metal oxide, plastic or another material. The structure can be a porous substrate and can also include a porous coating on some or all parts of the porous substrate, where the CO.sub.2-philic phase can be present in the pores of one or both of the porous substrate and the porous coating.

[0083] When the structure is comprised entirely of sorbent (e.g., CO.sub.2 binding molecules and a support), it can be formed by extrusion, molding, 3D printing, and the like, for example. The structure can be formed using support material without the CO_2-philic phase or using the support with the CO_2-philic phase already incorporated. When formed without using the CO_2-philic phase, the CO_2-philic phase can be incorporated into the structure through an impregnation, grafting, or other functionalization technique.

[0084] In a particular aspect, the support material can be applied to a substrate as a porous coating (also referred to as a "washcoat") on the surface of the substrate. In an embodiment, the porous coating can be a foam such as a polymeric foam (e.g., polyurethane foam, a polypropylene foam, a polyester foam, and the like), a metal foam, or a ceramic foam. The porous coating can include a metal-oxide layer (e.g., such as a foam). The metal-oxide layer can be silica or alumina on the surface of the substrate, for example. The porous coating can be present on the surface of the substrate, within the pores or voids of the substrate, or a combination thereof. The porous coating can be about 50 μm to 1500 μm thick and the pores can be of the dimension described above and herein.

[0085] In an aspect, the support material, the substrate, and/or the structure can be made of a ceramic substrate such as cordierite, alumina (e.g., γ-alumina, θ-alumina, δ-alumina), cordierite-a-alumina, silica, aluminosilicates, zirconia, germania, magnesia, titania, hafnia, silicon nitride, zircon mullite, spodumene, alumina-silica magnesia, zircon silicate, sillimanite, magnesium silicates, zircon, petalite, and combinations thereof. The metal or metal oxide structure can be aluminum, titanium, stainless steel, a Fe—Cr alloy, or a Cr—Al—Fe alloy. In cases where the oxide contains a formal charge, the charge can be balanced by appropriate counter-ions, such as cations of NR_4, Na, K, Ca, Mg, Li, H, Rb, Sr, Ba, Cs, or anions including phosphate, phosphite, sulfate, sulfate, nitrate, nitrite, chloride, bromide and the like.

[0086] In an aspect, the support material, the substrate, and/or the structure can be made of a plastic substrate that can be made of a polymer and/or copolymer of polyolefin(s), polyester(s), polyurethane(s), polycarbonate(s), polyetheretherketone(s), polyphenylene oxide(s), polyether sulfone(s), melamine(s), polyamide(s), polyvinylbenzene, polystyrene-divinylbenzene, polyurethane, polyacrylates, polystyrenes, polyacrylonitriles, polyimides, polyfurfural alcohol, phenol furfuryl alcohol, melamine formaldehydes, resorcinol formaldehydes, cresol formaldehyde, phenol formaldehyde, polyvinyl alcohol dialdehyde, polycyanurates, polyacrylamides, various epoxies, agar, and agarose, or combinations thereof.

[0087] In an embodiment, the structure can be a honeycomb structure such as a honeycomb monolith structure that includes channels. The honeycomb structure can have a regular, corrugated structure. The honeycomb monolith structure can have a length and width on the order of centimeters to meters while the thickness can be on the order of millimeters to centimeters or more. In an aspect, the honeycomb monolith structure does not have fibrous dimensions. In other words, the honeycomb structure can be a flow-through substrate comprising open channels defined by walls of the channels. The channels can have about 50 to about 900 cells per square inch. The channels can be polygonal (e.g., square, triangular, hexagonal, octagonal) sinusoidal, circular, or the like, in cross-section. Along the length of the channel, the channel length can have a configuration that is straight, zig-zag, skewed, or herringbone in shape. The length of the channel can be 1 mm to 10 s or 100 s of cm or more. The channels can have walls that are perforated or louvered. In an aspect, the sorbent can be disposed in the pores of the honeycomb structure and/or in the pores of a porous layer on the surface of the honeycomb structure. The honeycomb structure can have a geometric void fraction, otherwise known as the open face area, of between 0.3 to 0.95 or about 0.5 to 0.9.

[0088] In an embodiment, the honeycomb structure can comprise an inlet end, an outlet end, and inner channels extending from the inlet end to the outlet end. In some embodiments, the honeycomb comprises a multiplicity of cells extending from the inlet end to the outlet end, the cells being defined by intersecting cell or channel walls.

[0089] In an aspect, the honeycomb structure and/or substrate can be ceramic (e.g., of a type produced by Corning under the trademark Celcor®) that can be used with the sorbent in accordance with the principles of the present disclosure. The sorbent can be coated or otherwise immobilized on the inside of the pores of the ceramic honeycomb structure and/or within a porous layer on the surface of the ceramic honeycomb structure. In an aspect, the porous coating can include a metal-oxide layer such as silica or alumina on the surface of the substrate. In an embodiment, the metal-oxide layer can be mesoporous and macroporous. The honeycomb monolith may have a depth of 3 inches to 10 feet or about 3 and 24 inches.

[0090] In an aspect, the structure can be laminate sheets. Laminate sheets are structures containing a one-dimensional wall structure, whereby sheets are stacked upon one and other with space in between each sheet such that gas can flow between the sheets.

[0091] In an aspect, the structure can be a foam. Foams are structures with an irregular channel structure surrounded by an irregular solid structure. The solid structure is interconnected such that the foam material is self-standing.

[0092] In an aspect, the structure can be a plurality of fibers. Fibers are structures with high aspect ratio, and in gas contacting applications can be arranged in a regular array amongst one another when supported at least on one end of the fiber. The fibers can be solid or hollow.

[0093] In an aspect, the structure can be a minimal surface solid. Minimal surface solids are structures often used in packing for distillation and absorption systems to increase contact area with a material and a fluid. Minimal surface area solids are geometries that have zero mean surface area and include shapes such as gyroids. Gyroids can be sinusoidal, for example.

[0094] In an aspect, the structure can be a powder tray. Powder trays are structures whereby trays hold loose powder or pellets of the sorbent of the present disclosure to form a structured contactor without the material forming a self-standing structure by itself. Powder trays can be arranged in stacked layers to form sheets thereby forming a structure similar to a laminate. These layers can be created using flexible sheets, stiff sheets, or other flat surface that is mounted on a stiff frame structure. Powders are loose, free flowing solids with small characteristic particle diameter such as to provide a powdery consistency. Pellets are beads, balls, or other compacted structures used to provide structure and surface area to sorbents In an aspect, the structure can be sorbent particle volumes. Sorbent particle volumes can be contained by one or more walls such that gas can pass through them while keeping the sorbent contained. Sorbent particle volumes can be arranged relative to other sorbent particle volumes such as to approximate a honeycomb, fiber, or other structured contactor with a solid body.

[0095] In an aspect, the sorbent (e.g., structure) in the form of a contactor is an efficient embodiment for an effective method for capturing CO.sub.2 from ambient air or other gas mixtures (e.g., flue gas, exhaust gas, natural gas, or other gasses containing CO.sub.2) because a structured contactor, or a contactor, can be engineered to provide high surface area and low pressure drop for the air processing. Contactors can take the forms described of a honeycomb, a laminate sheet, a foam, fibers, minimal surface solids, powder trays, pellets, powder, and the like or a combination of two or more of the foregoing.

[0096] Now having described embodiments of the sorbent and structure, additional details regarding the systems and methods of the present disclosure are provided. The present disclosure provides for methods of capturing CO.sub.2 from ambient air or other gas mixtures (e.g., flue gas, exhaust gas, natural gas, or other gasses containing CO.sub.2). The method includes introducing the ambient air to the sorbent (e.g., structure), heating the sorbent (e.g., about 10 to 200° C. above the regular sorbent temperature to absorb the CO.sub.2) to at least a first temperature to controllably release the CO.sub.2; and collecting the CO.sub.2 in a CO.sub.2 collection device. The temperature increase in the sorbent can be performed by contacting the sorbent with a gas at elevated temperature, contacting the sorbent with a fluid at an elevated temperature, contacting the sorbent with a heat exchanger with hot fluid or gas running through it, by heating the walls of the container, vessel, or other containment device that contains the sorbent, or by contacting the sorbent with steam (e.g., the steam may be at a temperature between 60 to 200° C., and be saturated or superheated). In an aspect, the method can be implemented using the system described below.

[0097] The present disclosure provides for systems and devices for capturing CO.sub.2 from ambient air or other gas mixtures (e.g., flue gas, exhaust gas, natural gas, or other gasses containing CO.sub.2) where removal of CO.sub.2 is important. In general, the system includes a first device configured to introduce the ambient air or other gas mixture to the sorbent or contactor, where the sorbent or contactor includes those described herein. The sorbent is exposed to the ambient air or other gas mixture for a period of time (e.g., hours). In a particular aspect, the sorbent is a honeycomb monolith that has an open face area of between 0.3-0.95. The first device is configured to deliver the ambient air, for example, to the honeycomb monolith at a velocity of between 0.25-10 m/s. After the desired amount of time, a second device configured to heat the sorbent containing bound CO.sub.2 to at least a first temperature (e.g., about 40 to 200° C., about 50 to 200° C., or about 60 to 200° C.) to release the CO.sub.2. The second device of the system can operate to desorb CO.sub.2 by the sorbent. The second device can include components to support temperature swing, pressure swing, steam swing, concentration swing, combinations thereof, or other dynamic processes to desorb the CO.sub.2. In an embodiment, the steam swing process can include exposing the sorbent to steam, where the temperature of the steam is about 60° C. to 150° C. and the pressure of the steam is about 0.2 bara to 5 bara. A third device is configured to collect the released CO.sub.2. The system can be operated so that the sorbent absorbs and desorbs the CO.sub.2 in an efficient and cost-effective manner.

[0098] Aspects of the present disclosure can be described by the following paragraphs.

[0099] The present disclosure provides for a sorbent comprising: a CO.sub.2-philic phase and a support, wherein the CO.sub.2-philic phase includes the reaction product of a substituted epoxide and an amine. The amine is an amine polymer. The amine polymer can be branched, hyperbranched, dendritic, or linear. The amine polymer is one of polyethylenimine, polypropylenimine, polyallylamine, polyvinylamine, polyglycidylamine, or polystyrene-divinylbenzene polymer functionalized with amine, or other amine polymers. The CO.sub.2-philic phase homogeneous. The CO.sub.2-philic phase heterogeneous. The fraction of amines modified by reaction with a substituted epoxide is about 0.001 to 1 of total primary and secondary amines in the amine polymer, where a fraction of 1 means all of the primary and secondary amines. The fraction of amines modified by reaction with a substituted epoxide is about 0.01 to 0.5 of total primary and secondary amines in the amine polymer, where a fraction of 1 means all of the primary and secondary amines. The amine polymer is physically impregnated into pores of the support. The amine polymer is physically impregnated onto the surface of the support. The amine polymer is covalently bonded to the surface of the support. The substituted epoxide is 1,2-epoxy-2-methylpropane, 2,3-epoxybutane, 3,4-epoxy-1-butene, 3,3-dimethyl-1,2-epoxybutane, 2-methyl-2-vinyloxirane, 1,2-epoxy-3,3,3-trifluoropropane, 2-phenylpropylene oxide, 1-(2,3-epoxypropyl)-2-nitroimidazole, trans-stilbene oxide, cis-stilbene oxide, a mixture of trans- and cis-stilbene oxide, 1,2-epoxy-5-hexene, epoxystyrene, (2,3-epoxypropyl)benzene, 2-(4-chlorophenyl) oxirane, 2-(4-bromophenyl) oxirane, 2-(4-fluorophenyl) oxirane, cyclohexene oxide, 4-vinyl-1-cyclohexene-1,2-epoxide, or 1,3-butadiene diepoxide. The CO.sub.2-philic phase includes a structure selected from at least one of the following structures, where R.sub.x is the substituted group:

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wherein R.sub.1, R.sub.2, R.sub.3, and R.sub.4 are each independently selected from a hydrogen atom, an alkyl group, an allyl group, an alkoxy group, an alkylhalide group, an aryl group, a benzyl group, a phenyl group, a phenol group, a heteroaryl group, a nitroimidazole group and the like, and combinations thereof (each group can independently be substituted or unsubstituted), and wherein each R′ are independently selected from a hydrogen atom, an alkyl group, an alkoxy group, an alkylhalide group, an aryl group, a benzyl group, a phenyl group, a phenol group, an amine group (e.g., alkylenimine (C2 to C8) such as ethylenimine and propylenimine), a heteroaryl group, a nitroimidazole group, an alkyl amino group, and combinations thereof (each group can independently be substituted or unsubstituted). The support is ceramic, metal, metal oxide, plastic, cellulose, carbon, a zeolite, a metal organic framework (MOF), a porous organic framework (POF), a covenant organic framework (COF), a polymer of intrinsic microporosity (PIM), a polymer, a fibrous cellulose, fiberglass, or boron-nitride fiber.

[0100] The present disclosure provides for a contactor, comprising a structure and the sorbent as described above and herein. The structure is selected from a honeycomb, a laminate sheet, a foam, fibers, a minimal surface solid, powder trays, pellets, or a combination of these.

[0101] The present disclosure provides for a system for capturing CO.sub.2 from a gas, optionally the gas is ambient air, comprising: a first device configured to introduce the gas to the sorbent or contactor as described and herein to bind CO.sub.2 to the sorbent; a second device configured to heat the sorbent containing bound CO.sub.2 to at least a first temperature to release the CO.sub.2; and a third device configured to collect the released CO.sub.2. After being heated the sorbent is regenerated so it able to absorb CO.sub.2 from the gas. The sorbent is in the form of a honeycomb, a laminate sheet, a foam, fibers, a minimal surface solid, powder trays, pellets, a combination thereof. The honeycomb has an open face area of between 0.3-0.95. The gas approaches the honeycomb at a velocity of between 0.25-10 m/s. The system is configured to operate to remove CO.sub.2 from ambient air, where the ambient air has a low concentration of CO.sub.2.

[0102] The present disclosure provides for a method of capturing CO.sub.2 from gas optionally the gas is ambient air comprising: introducing the ambient air to the sorbent as described above and herein to bind CO.sub.2 to the sorbent; heating the sorbent to at least a first temperature to controllably release the CO.sub.2; and collecting the CO.sub.2 in a CO.sub.2 collection device. Heating the sorbent regenerates the sorbent so it is able to absorb CO.sub.2 from ambient air. The sorbent is heated by contacting it with steam. The method is configured to operate to remove CO.sub.2 from ambient air, where the ambient air has a low concentration of CO.sub.2. The sorbent is in the form of a honeycomb, a laminate sheet, a foam, fibers, a minimal surface solid, powder trays, pellets, a combination thereof.

[0103] The present disclosure provides for a system for implement the method as described above and herein.

EXAMPLES

[0104] The removal of CO.sub.2 from ambient air through engineered chemical processes, otherwise known as Direct Air Capture (DAC), is emerging as an important environment technology for the mitigation of climate change. DAC is a technology that can provide negative emissions, removing CO.sub.2 from the atmosphere. However, the current DAC technology is expensive thereby limiting its deployment. Therefore, improvements to DAC technology are needed. Many DAC technologies rely on solid sorbent materials as a medium to perform the separation of CO.sub.2 from the air. These sorbents are generally applied in temperature swing processes, where at low temperature CO.sub.2 from the air binds to the sites within them, and then at high temperature the CO.sub.2 is released into a concentrated product that can be sequestered or sold as a product. Many DAC sorbents utilize amines to bind CO.sub.2 in this manner. Certain amine types can be effective at binding CO.sub.2 from low concentrations such as that found in the air (400 ppm).

[0105] While some amine types are effective at binding CO.sub.2 from ambient air, they slowly oxidize in air from ambient oxygen. This effect is exacerbated in process cycles that raise the temperature of the sorbent to remove bound CO.sub.2, thereby creating accelerated oxidative degradation that reduces the lifetime of the CO.sub.2 sorbents. Therefore, sorbents with improved oxidative stability that are effective at removing CO.sub.2 from ambient air are needed.

[0106] Some sorbents used in DAC processes are composite materials, containing a CO.sub.2-philic phase (e.g., CO.sub.2 binding molecules and and phenol containing molecules) that is distributed in or within a solid material that provides it with surface area. The CO_2-philic phase can be grafted to the solid surface, physically impregnated into the pores of the solid material, or physically supported on the surface of the solid material. The CO_2-philic phase of these sorbents can be amines or other molecules that can bind CO_2. In some cases, the amines can be polymeric amines such as polyethylenimine, polypropylenimine, polyallylamine, polyvinylamine, polybutylamine or others. These polymeric amines can be linear, branched, hyperbranched, dendritic, or take the form some other macromolecule. In other cases, the amines can be small molecules such as TEPA, TPTA or others. In other cases, the amines can be aminosilanes. The solid support material can be a metal oxide, carbon, metal, or other structure that can provide ample surface area for the CO_2-philic phase to be deposited to allow for useful CO_2 adsorption and desorption capacities and kinetics. In this way, the solid support material is functionalized with the CO_2-philic phase to create a composite sorbent.

[0107] The sorbents can be formed or incorporated into macrostructures, or contactors, to provide advantages in applications such as DAC. Such structures can be honeycomb monoliths, laminate sheets, pellets, or other structures that can provide a high geometric surface area for air or CO_2 containing gasses to efficiently contact the sorbent such that the CO_2 can bind to the sorbent.

[0108] The sorbents can be utilized in processes to capture CO_2 from air or a variety of other gas stream such as flue gas, natural gas and others. These processes are known as "CO_2 Capture Processes". CO_2 capture processes can utilize temperature swing, concentration swing, pressure swing, steam stripping or other swing techniques to remove CO_2 that has been bound the surface of the sorbent.

[0109] There has been relatively little development of improved CO_2 sorbents utilizing polyethylenimine, and especially for DAC applications.

[0110] Now having described the embodiments of the present disclosure, in general, the following examples describes some additional embodiments of the present disclosure. While embodiments of the present disclosure are described in connection with these examples and the corresponding text and figures, there is no intent to limit embodiments of the present disclosure to these descriptions. On the contrary, the intent is to cover all alternatives, modifications, and equivalents included within the spirit and scope of embodiments of the present disclosure.

Example: Schematic Showing Sorbent Having a Support Material and CO_2-Philic Phase

[0111] FIG. **1**B shows the primary components of the sorbent system. The sorbent system is comprised of a support material and a CO_2-philic phase. In the schematic shown, a single CO_2-philic phase is shown incorporated into a single support. The CO_2-philic phase shown is polyethylenimine (PEI). One known adsorption product of CO_2 and PEI is ammonium carbamate. Mesoporous alumina is shown as a support.

Example: Schematic of Honeycomb Monolith Structure Containing a Sorbent Washcoat

[0112] FIG. **2** shows one embodiment of a honeycomb monolith contactor. The figure shows the primary geometrical features of a honeycomb monolith, having straight, flow-through channels surrounded on all sides by walls. The figure schematically shows a washcoat applied to the walls, where the washcoat is comprised of sorbent. The sorbent is comprised of a support material and a CO_2-philic phase, such as PEI.

Example: Substituted Epoxides Evaluated as Modifiers to Polyethylenimine (PEI)

[0113] A variety of substituted epoxides were evaluated to assess their ability to reduce the oxidation rate when reacted with PEI and incorporated into a support. The substituted epoxides and their reference name are provided in FIG. **3**.

Example: Preparation of Improved CO_2 Sorbents Containing Polyethylenimine (PEI) Reacted with Substituted Epoxides

[0114] Branched polyethylenimine (PEI) MW 800 is available from Sigma Aldrich. The substituted epoxides used are listed in the table below. The PEI was dissolved in a solution of methanol, after which a single epoxide was added in a ratio corresponding to 0.5 mol epoxide per mol of primary amine present in PEI. The mixture was allowed to stir at room temperature for a about 6 to 24 hours or about 12 to 24 hours to allow the epoxide to react with the PEI. Separately, a mesoporous alumina was dispersed in methanol and mixed until homogeneous. Then, the epoxide functionalized PEI solution was pipetted into the alumina/methanol dispersion. After stirring for >3 h, the solvent was removed by rotary evaporation and subsequent drying in a vacuum oven at 100° C. for 10 h. Mass ratios of the alumina and epoxide functionalized PEI were controlled such as to achieve 40-80% filling of the mesopores of the mesoporous alumina with epoxide functionalized PEI. The resultant composite sorbents were of a powdery consistency.

TABLE-US-00001 Substituted Epoxide CAS Vendor Epoxystyrene (Epoxide 5) 96-09-3 Sigma Aldrich 3,3-dimethyl-1,2-epoxybutane (Epoxide 6) 2245-30-9 Sigma Aldrich 2-(4-fluorophenyl) oxirane (Epoxide 7) 18511-62-1 Sigma Aldrich 2-(4-chlorophenyl) oxirane (Epoxide 8) 2788-86-5 Sigma Aldrich cyclohexene oxide (Epoxide 9) 286-20-4 Sigma Aldrich 1-(2,3-epoxypropyl)-2-nitroimidazole 13551-90-1 Sigma Aldrich (Epoxide 10) 3,4-epoxy-1-butene (Epoxide 11) 930-22-3 Sigma Aldrich trans-stilbene oxide (Epoxide 12) 1439-07-2 Sigma Aldrich

Example: Characterization of Improved CO.SUB.2.-Philic Phases and CO.SUB.2 .Sorbents

[0115] Chemical characterization was carried out to confirm the properties of the improved CO.sub.2-philic phases. Further chemical characterization was carried out to confirm that these CO.sub.2-philic phases were successfully incorporated into the pores of a mesoporous support to create a CO.sub.2 sorbent.

[0116] 1H NMR experiments were carried out on PEI and the materials resulting after reaction of PEI with various substituted epoxides to characterize the nature of the resultant material.

[0117] TGA burnoff experiments were carried out on sorbents comprised of the improved CO.sub.2-philic phases to characterize the total quantity of organic present in the sorbent. Samples were heated under diluted air to 900° C. and their mass loss tracked. Total organic content was taken as the mass loss over that temperature interval, after removing the contribution of CO.sub.2 and H.sub.2O lost at lower temperatures.

Example: TGA Burnoff Experiments

[0118] FIGS. **4** and **5** shows the mass loss and heat flow (DSC) curves during exposure of materials to diluted air while ramping the temperature from room temperature to 900° C. The figures show that the native PEI and modified PEI sorbents both lose substantial mass during the experiment, as the organic fractions of the materials burn off from the oxidative conditions. The DSC traces reveal different oxidative profiles for the materials during the course of the exposure and also confirm that the PEI has been modified by reaction.

Example: Testing in CO.SUB.2 .Adsorption Processes

[0119] Sorbents created with improved CO.sub.2-philic phases were tested for CO.sub.2 adsorption in a TGA under 400 ppm CO.sub.2 at 30° C. to simulate the gas contacting step of a Direct Air Capture process. The sorbents were first treated in N.sub.2 at 100° C. to desorb any bound H.sub.2O and CO.sub.2 before being equilibrated at 30° C. under N.sub.2. Then, the gas concentration was switched isothermally to contain 400 ppm CO.sub.2 balanced by N.sub.2 and the mass change was recorded. Under these moisture free conditions, the mass gain of the material corresponds to the adsorption of CO.sub.2 and therefore can be used to measure the total quantity and rate of CO.sub.2 adsorption onto the materials.

[0120] Various improved sorbents were characterized for CO.sub.2 adsorption using this method, and compared to the baseline PEI based sorbent. Sorbents created using PEI reacted with a variety of substituted epoxides at a loading of 0.5 mol substituted epoxide/mol primary N in PEI were evaluated relative to the sorbent with PEI alone.

[0121] The sorbents were evaluated on two bases, i) CO.sub.2 capacity, mmol of CO.sub.2 adsorbed per mol of sorbent present, and ii) amine efficiency, mmol of CO.sub.2 adsorbed per mol of N in PEI present in sample. The former unit of performance is useful to show bulk sorbent performance, and the second unit of performance is useful to evaluate the performance of the amine polymer itself and takes into account changes to the bulk composition of the sorbent.

Example: CO_2 Adsorption Uptake Capacities at 400 ppm CO_2 of PEI and improved CO_2-Philic Phases Supported on Mesoporous Alumina

[0122] FIG. **6** shows transient TGA uptake curves under DAC conditions for sorbents created with improved CO_2-philic phases utilizing PEI reacted with substituted epoxides, labeled by their reference in FIG. **3**, each at 0.5 mol substituted epoxide/mol primary N in PEI, compared to that of unmodified PEI. Each of the sorbents are able to adsorb CO_2 at the ultra dilute conditions of 400 ppm, with varying degrees of amine efficiency. Reaction with Epoxide 11, Epoxide 9, Epoxide 6, and Epoxide 7 appear to not impact the ability of the material to adsorb CO_2 compared to PEI. Reaction with Epoxide 5, Epoxide 8, and Epoxide 12 reduces the ability to adsorb CO_2 to some extent, reaction with Epoxide 10 appears to result in the loss of almost all ability to adsorb CO_2. This is an unexpected result, as the epoxidation of amines converts primary amines to secondary amines and secondary amines to tertiary amines. Secondary amines are thought to have a less favorable interaction potential for CO_2 compared to primary amines due to the steric hindrance of an additional group bonded to them. Tertiary amines are well known to not interact with CO_2 under dry conditions such as those tested here. Therefore, this result shows that the epoxidation of PEI unexpectedly does not usually lead to substantial loss in performance of PEI as a CO_2-philic phase. Additionally, reaction with the substituted epoxides either improve or maintain the initial rate of adsorption of CO_2 as compared to PEI, with the exception of Epoxide 10. The initial rate of adsorption follows the trend of PEI-0.5 Epoxide 9≈PEI-0.5 Epoxide 11≈PEI-0.5 Epoxide 6≈PEI-0.5 Epoxide 7>PEI-0.5 Epoxide 8≈PEI-0.5 Epoxide 5≈PEI≈PEI-0.5 Epoxide 12>PEI-0.5 Epoxide 10. This indicates that the epoxidation of PEI with substituted epoxides can improve the CO_2 adsorption rate, thereby allowing sorbents to equilibrate more quickly when adsorbing CO_2 from air or other dilute streams. This unexpected result may be explained by the epoxide side chains creating void spaces within the CO_2-philic phase that allow unbound CO_2 to more readily diffuse into the CO_2-philic phase compared to the unmodified PEI.

Example: Testing of Oxidative Stability

[0123] The oxidative stability of materials can be probed in several ways In this study, the oxidative stability of the sorbents was evaluated by tracking the heat flow evolved from the materials using a DSC during exposure to isothermal oxidative conditions. Here, the sorbents were first treated in inert gas at 100° C. to desorb any bound H_2O and CO_2 before being equilibrated at either 125, 137.5 or 150° C. under inert gas for 60 minutes. The gas was then isothermally switched to a 17% O_2 mixture (CO_2-free air, or air) and held until the reaction finished. This isothermal, oxidative environment was maintained for a specific amount of time to measure the heat flow and mass loss. To prevent any further oxidation, the sample was then cooled under N_2 to room temperature. During these experiments, for each oxidative condition, the DSC measures the incremental heat flux, which increases, levels out, and then decreases to zero. The oxidation was considered complete when the integrated heat flow over 10 min changed less than +0.01% of the total integrated heat. To determine the extent of oxidation as a function of time, DSC data were converted from the base unit of mW/mg sorbent to W/gPEI using the PEI loading measured by TGA burnoff. DSC data were corrected for drift by applying an offset, determined by the heat flow value when the DSC curve approached a horizontal line. The total heat evolved was calculated by integrating heat flow over time. The extent of oxidation from DSC was calculated by dividing the integral heat flow curve by the total heat evolved. This method has been previously calibrated with the loss in amine efficiency as being a method of tracking the chemical reaction rate of oxidative degradation in-situ, and is shown in FIGS. **7**A-**7**D. Further details on this method and its validation are discussed in the following papers: Nezam et al, ACS Sustainable Chem. Eng., 2021, 9, 8477-8486, and Racicot et al, J. Phys. Chem. C, 2022, 126, 8807-8816, which is incorporated herein by reference.

[0124] FIG. **8** shows oxidation curves in the presence of CO.sub.2-free air for CO.sub.2 sorbents created with PEI and PEI reacted with substituted epoxides, labeled by their reference in FIG. **3**, at a ratio of 0.5 mol substituted epoxide/mol primary N in PEI, and supported in mesoporous alumina. The figure shows two general categories of behavior; some substituted epoxides accelerate the oxidation of PEI, while the majority of substituted epoxides slow the oxidation rate of PEI, though to varying extents. In particular, Epoxide 5, Epoxide 6, Epoxide 7, Epoxide 8, Epoxide 9, Epoxide 11, and Epoxide 12 slows the rate of oxidation when reacted with PEI, while Epoxide 10 accelerates the rate of oxidation when reacted with PEI. The differences between the extent of oxidation observed for the different substituted epoxides in FIG. **8** suggest that not all substituted epoxides will improve, or reduce, the rate of oxidation, and some can in fact accelerate the rate of oxidation as compared to PEI, as is the case for Epoxide 10 (1-(2,3-epoxypropyl)-2-nitroimidazole).
[0125] Example structures of modified amines on modified amine polymers are illustrated in FIG. **9**.
[0126] Example chemical structures of 1-amino-2-alcohol groups reacted with a primary amine to form a secondary amine are illustrated in FIG. **10**.
[0127] It should be noted that ratios, concentrations, amounts, and other numerical data may be expressed herein in a range format. It is to be understood that such a range format is used for convenience and brevity, and thus, should be interpreted in a flexible manner to include not only the numerical values explicitly recited as the limits of the range, but also to include all the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range is explicitly recited. To illustrate, a concentration range of “about 0.1% to about 5%” should be interpreted to include not only the explicitly recited concentration of about 0.1 wt % to about 5 wt %, but also include individual concentrations (e.g., 1%, 2%, 3%, and 4%) and the sub-ranges (e.g., 0.5%, 1.1%, 2.2%, 3.3%, and 4.4%) within the indicated range. In an aspect, the term “about” can include traditional rounding according to significant figures of the numerical value. In addition, the phrase “about ‘x’ to ‘y” includes “about ‘x’ to about ‘y’”.
[0128] Many variations and modifications may be made to the above-described aspects. All such modifications and variations are intended to be included herein within the scope of this disclosure and protected by the following claims.

Claims

1. A sorbent comprising: a CO.sub.2-philic phase and a support, wherein the CO.sub.2-philic phase includes the reaction product of a substituted epoxide and an amine polymer.
2. (canceled)
3. The sorbent of claim 1, wherein the amine polymer is branched, hyperbranched, dendritic, or linear.
4. The sorbent of claim 3, wherein the amine polymer is one of polyethylenimine, polypropylenimine, polyallylamine, polyvinylamine, polyglycidylamine, or polystyrene-divinylbenzene polymer functionalized with amine.
5. The sorbent of claim 1, wherein the CO.sub.2-philic phase is homogeneous.
6. The sorbent of claim 1, wherein the CO.sub.2-philic phase is heterogeneous.
7. The sorbent of claim 1, wherein a fraction of amines reacted with the substituted epoxide is about 0.001 to 1 or is about 0.01 to 0.5 of total primary and secondary amines in the amine polymer, where a fraction of 1 means all of the primary and secondary amines are reacted.
8. (canceled)
9. The sorbent of claim 1, wherein the amine polymer is physically impregnated into pores of the support, or is physically impregnated onto the surface of the support.
10. (canceled)

11. The sorbent of claim 1, wherein the amine polymer is covalently bonded to the surface of the support.
12. The sorbent of claim 1, wherein the substituted epoxide is 1,2-epoxy-2-methylpropane, 2,3-epoxybutane, 3,4-epoxy-1-butene, 3,3-dimethyl-1,2-epoxybutane, 2-methyl-2-vinyloxirane, 1,2-epoxy-3,3,3-trifluoropropane, 2-phenylpropylene oxide, 1-(2,3-epoxypropyl)-2-nitroimidazole, trans-stilbene oxide, cis-stilbene oxide, a mixture of trans- and cis-stilbene oxide, 1,2-epoxy-5-hexene, epoxystyrene, (2,3-epoxypropyl)benzene, 2-(4-chlorophenyl) oxirane, 2-(4-bromophenyl) oxirane, 2-(4-fluorophenyl) oxirane, cyclohexene oxide, 4-vinyl-1-cyclohexene-1,2-epoxide, or 1,3-butadiene diepoxide.
13. The sorbent of claim 1, wherein the CO.sub.2-philic phase includes a structure selected from at least one of the following structures, where R.sub.x is the substituted group: ##STR00005## wherein R.sub.1, R.sub.2, R.sub.3 and R.sub.4 are each independently selected from a hydrogen atom, an alkyl group, an allyl group, an alkoxy group, an alkylhalide group, an aryl group, a benzyl group, a phenyl group, a phenol group, a heteroaryl group, a nitroimidazole group and the like, and combinations thereof, and wherein each R′ are independently selected from a hydrogen atom, an alkyl group, an alkoxy group, an alkylhalide group, an aryl group, a benzyl group, a phenyl group, a phenol group, an amine group, a heteroaryl group, a nitroimidazole group, an alkyl amino group, and combinations thereof.
14. The sorbent of claim 1, wherein the support is ceramic, metal, metal oxide, plastic, cellulose, carbon, a zeolite, a metal organic framework (MOF), a porous organic framework (POF), a covenant organic framework (COF), a polymer of intrinsic microporosity (PIM), a polymer, a fibrous cellulose, fiberglass, or boron-nitride fiber.
15. A contactor comprising: a structure and a sorbent, wherein the sorbent includes a CO.sub.2-philic phase and a support, wherein the CO.sub.2-philic phase includes the reaction product of a substituted epoxide and an amine polymer.
16. The contactor of claim 15, wherein the structure is selected from a honeycomb, a laminate sheet, a foam, fibers, a minimal surface solid, powder trays, pellets, or a combination of these.
17. A system for capturing CO.sub.2 from a gas, the system comprising: a first device configured to introduce the gas to a sorbent to bind CO.sub.2 to the sorbent, wherein the sorbent includes a CO.sub.2-philic phase and a support, and the CO.sub.2-philic phase includes the reaction product of a substituted epoxide and an amine polymer; a second device configured to heat the sorbent containing bound CO.sub.2 to at least a first temperature to release the CO.sub.2; and a third device configured to collect the released CO.sub.2.
18-22. (canceled)
23. A method of capturing CO.sub.2 from gas, the method comprising: introducing the ambient air to the sorbent of claim 1 to bind CO.sub.2 to the sorbent; heating the sorbent to at least a first temperature to controllably release the CO.sub.2; and collecting the CO.sub.2 in a CO.sub.2 collection device.
24-28. (canceled)
29. The sorbent of claim 1, wherein the substituted epoxide is 2-(4-chlorophenyl) oxirane.
30. The sorbent of claim 1, wherein the amine polymer is polyethyleneimine.
31. The contactor of claim 15, wherein the structure is a honeycomb.
32. The system of claim 17, wherein the gas is ambient air.
33. The method of claim 23, wherein the gas is ambient air.
